

## 5. Frequency Domain High-Pass Filtering

*No specific frequency values were supplied, so the sample set below (ordered from lowest to highest frequency) is used.*

| Component | F1 | F2 | F3 | F4 | F5  | F6  |
|-----------|----|----|----|----|-----|-----|
| Magnitude | 10 | 15 | 20 | 90 | 130 | 150 |

### (a) Interpretation of high-frequency importance

The last three components (F4, F5, F6 = 90, 130, 150) have much larger magnitudes and represent high-frequency content — these correspond to edges, fine texture, and sharp intensity transitions in the image. The first three components (F1–F3 = 10, 15, 20) are small, low-frequency values representing the smooth, slowly-varying background.

### (b) Justification for high-pass filtering

Since the meaningful detail (edges/texture) is concentrated in the high-frequency components, a high-pass filter should be applied to retain and enhance F4–F6 while suppressing the low-frequency background (F1–F3), sharpening the image and emphasising object boundaries.

### (c) Apply filtering (remove low frequencies)

F1, F2, F3 are set to 0; F4, F5, F6 are retained unchanged.

### (d) Computed output

| Component | F1 | F2 | F3 | F4 | F5  | F6  |
|-----------|----|----|----|----|-----|-----|
| Filtered  | 0  | 0  | 0  | 90 | 130 | 150 |

### (e) Interpretation of result

The filtered spectrum = [0, 0, 0, 90, 130, 150]. With the low-frequency background removed, reconstructing the image from this spectrum would emphasise edges and fine details, producing a sharpened, high-contrast version of the original with the smooth shading removed.